

# **Perceptron vs neuron, Single layer Perceptron and Multi Layer Perceptron**

# 1. Perceptron: The Simplest Neural Unit

- A **perceptron** is the earliest form of a neural network unit, introduced by **Frank Rosenblatt** in 1958. It is a **binary classifier** that makes predictions based on a linear combination of input features.
- The perceptron algorithm was one of the first algorithms used to implement a simple neural network.
- **Perceptrons are supervised learning algorithms and is a type of an ANN.**

# Components of a Perceptron:

- **Inputs:** The perceptron takes several inputs ( $x_1, x_2, \dots, x_n$ )
- **Weights:** Each input is associated with a weight ( $w_1, w_2, \dots, w_n$ ).
- **Bias:** A bias term ( $b$ ) is added to shift the decision boundary.
- **Activation Function:** The perceptron uses a **step function** (a simple thresholding function) to determine whether the weighted sum of inputs plus the bias is above or below a certain threshold.
- The mathematical representation is:

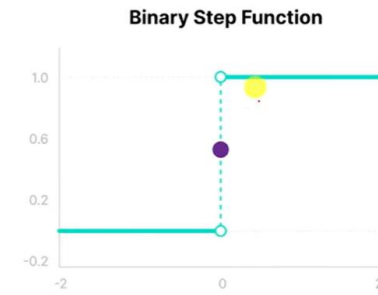
$$y = \begin{cases} 1 & \text{if } \sum_{i=1}^n w_i x_i + b > 0 \\ 0 & \text{if } \sum_{i=1}^n w_i x_i + b \leq 0 \end{cases}$$

- **Binary Output:** The output of a perceptron is binary (1 or 0), making it suitable for linearly separable classification problems.

# Activation function

Binary step

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$



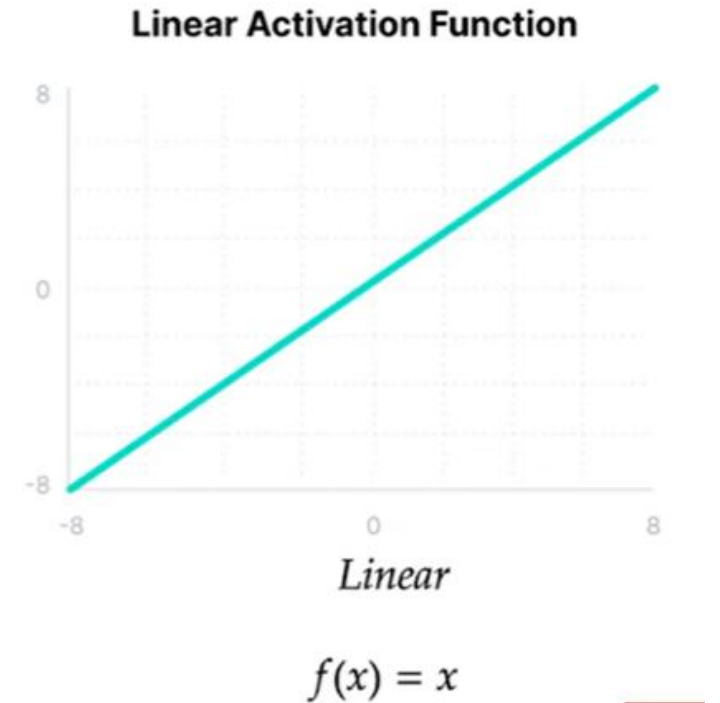
- Activation function decides whether the neuron should be activated or not.
- This means that it will decide whether the neuron's input to the network is important or not in the process of prediction using simpler mathematical operations.

## Binary Step Function:

- It depends on a threshold value that decides whether the neuron should be activated or not.
- The weighted sum is fed to the activation function and is compared to certain threshold; if the weighted sum is greater than it, then the neuron is activated, else it is deactivated, meaning that its output is not passed on to the next hidden layer.
- It cannot be used for multivalued output. That is multi class classification.
- Gradient of step function is zero, which causes a hindrance in the backpropagation process. So gradient of the step function should be greater than one.

# Linear Activation function

- It is also called as no activation or identity function. That is the activation is proportional to the input.
- It is not possible to use in backpropagation as a derivative of the function is a constant and has no relation to the input  $x$ .
- All layers of the neural network will collapse into one if we use linear activation function. It linear network into just one layer. That is the last layer will be linear function of first layer.



# Non Linear Activation function

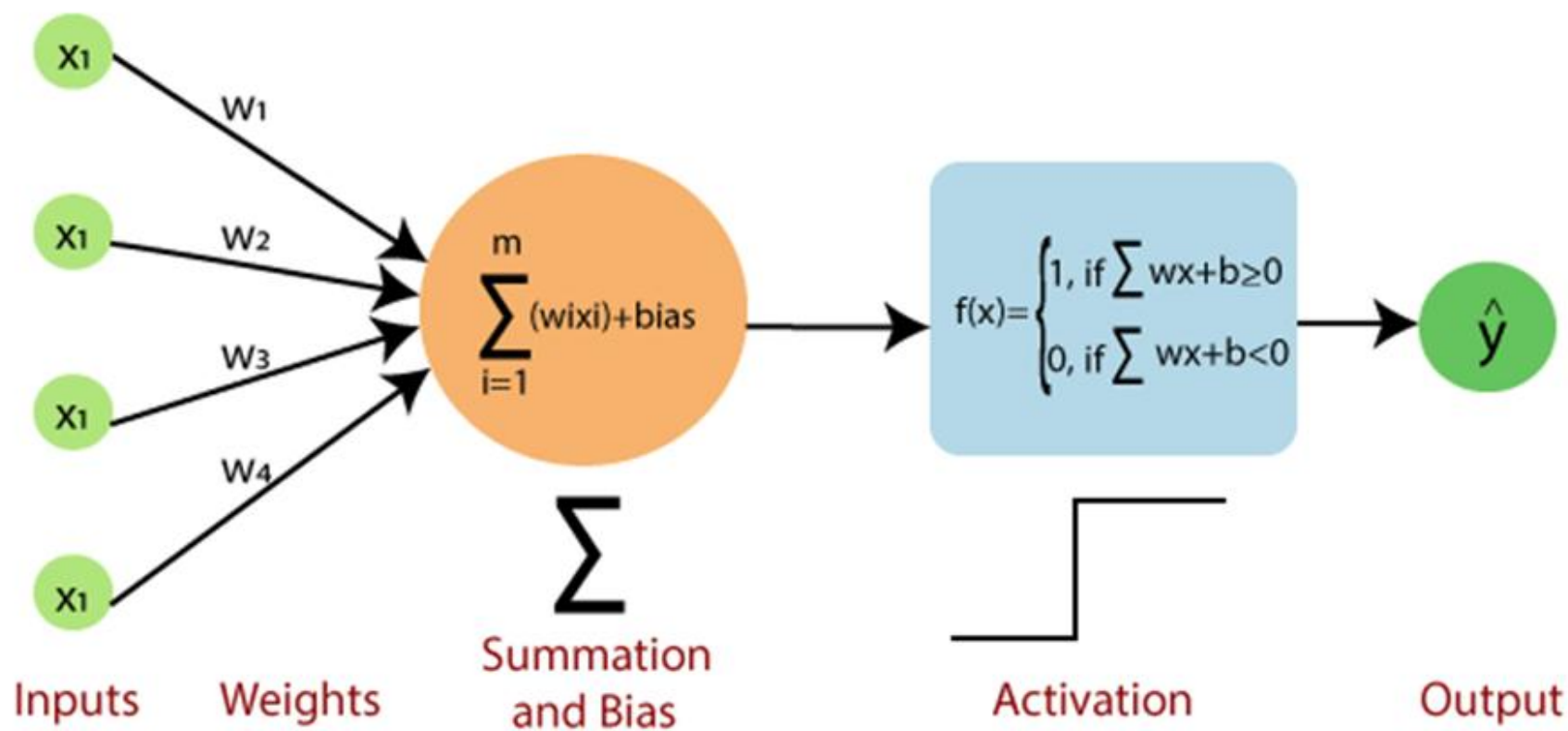
- It can be used in backpropagation because now the derivative function will be related to the input, and it is possible to go back and understand which weights in the input neuron can provide better prediction.
- They allow stacking of multiple layers of neurons as the output would now be a non-linear combination of input passed through multiple layers.

# Non-linear activation function

- Sigmoid
- Tanh
- ReLU
- Leaky ReLU
- Parametric ReLU
- ELU
- Softmax
- Swish
- GELU
- SELU

# Limitations of Perceptron:

- **Linear separability:** The perceptron can only solve problems that are linearly separable (i.e., it can only classify data that can be separated by a straight line or hyperplane). It cannot solve more complex problems like XOR.
- **Single-layer model:** The original perceptron is a single-layer model and does not have hidden layers, limiting its expressiveness.



## 2. Neuron: A Generalized Unit in Neural Networks

- A **neuron**, or artificial neuron, is a more generalized version of the perceptron and is the building block of modern deep learning architectures. Neurons in deep learning are part of **multi-layer neural networks**, which can have multiple hidden layers.

# Key Differences and Features of a Neuron:

- **Activation Function:** Unlike the perceptron, which uses a simple step function for activation, neurons in modern neural networks can use a variety of activation functions, such as:

- Sigmoid:  $\sigma(x) = \frac{1}{1+e^{-x}}$
- Tanh:  $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- ReLU (Rectified Linear Unit):  $\text{ReLU}(x) = \max(0, x)$  These activation functions introduce non-linearity, allowing the network to model complex patterns.

- **Multi-layer Networks:** Neurons are part of more sophisticated architectures called **multi-layer perceptrons (MLPs)** or deep neural networks, where neurons are organized into layers (input layer, hidden layers, and output layer). Each layer performs computations, and the output of one layer is fed as input to the next.
- **Continuous Output:** Neurons can output continuous values, unlike the binary output of a perceptron. This makes them more versatile for tasks like regression, multi-class classification, and complex feature extraction.
- **Learning through Backpropagation:** Neurons in deep learning models are trained using **backpropagation** and gradient descent, which adjusts the weights based on the error between the predicted and actual outputs. The perceptron uses a simpler update rule that works only for linearly separable problems.

# Modern Neural Networks:

- **Deep Learning Neurons:** In modern deep learning, neurons are the generalized units that can handle non-linear patterns and are stacked in multiple layers, giving rise to **deep neural networks (DNNs)**.
- **Non-linearity:** The introduction of non-linear activation functions allows neurons to model more complex, non-linearly separable data, overcoming the limitations of the simple perceptron

# Key Differences Between Perceptron and Neuron in Deep Learning

| Aspect              | Perceptron                                  | Neuron in Deep Learning                                       |
|---------------------|---------------------------------------------|---------------------------------------------------------------|
| Output              | Binary (0 or 1)                             | Continuous or non-linear (depends on the activation function) |
| Activation Function | Step function                               | Sigmoid, ReLU, Tanh, Softmax, etc.                            |
| Complexity          | Simple, handles linearly separable problems | Complex, handles non-linear, multi-dimensional problems       |
| Layers              | Single layer (original perceptron)          | Multi-layer (input, hidden, and output layers)                |
| Learning            | Basic learning rule for linear problems     | Gradient descent with backpropagation                         |
| Use Case            | Binary classification                       | Classification, regression, and various deep learning tasks   |

# Comparison Table

| Aspect              | Single-Layer Perceptron (SLP)                        | Multi-Layer Perceptron (MLP)                                    |
|---------------------|------------------------------------------------------|-----------------------------------------------------------------|
| Architecture        | One input layer, one output layer (no hidden layers) | Input, hidden, and output layers                                |
| Problem Solvability | Solves only linearly separable problems (e.g., AND)  | Solves both linear and non-linear problems (e.g., XOR)          |
| Activation Function | Step function (binary output)                        | Non-linear functions (e.g., sigmoid, ReLU, tanh)                |
| Learning Algorithm  | Perceptron learning rule (no backpropagation)        | Trained using backpropagation and gradient descent              |
| Output              | Binary (0 or 1)                                      | Continuous (for regression) or multi-class (for classification) |
| Applications        | Simple binary classification                         | Complex tasks (classification, regression, image recognition)   |
| Complexity          | Simple and limited                                   | Complex and powerful                                            |